



HIGH SCHOOL · STUDENT EDITION

# AI Innovators

## Student Design & Decision Portfolio

Investigate real problems. Verify the evidence. Design responsible solutions.  
Defend the decisions only people can make.



Healthcare



Construction & Trades



Technology & Cyber



Drone / UAS



Business



Education & Media

**PORTFOLIO OWNER**

Name or classroom code

Team

Period

\_\_\_\_\_

AI assists. People verify.  
People decide. People remain  
accountable.

YOUR ARC

Understand

Evaluate

Analyze

Protect

Apply & Create

## ORIENTATION · AI INNOVATORS PROFESSIONAL ROUTINE

# Pause → Protect → Prompt → Prove → Preserve Human Judgment

Tell It → Limit It → Check It → People Decide stays your core Future Ready AI framework. As an AI Innovator, you'll also use this professional routine—the fuller version people use at work when decisions carry real consequences.

|   |                                |                                                                                                |
|---|--------------------------------|------------------------------------------------------------------------------------------------|
| 1 | <b>Pause</b>                   | Should AI be used here at all? What is at stake?                                               |
| 2 | <b>Protect</b>                 | What information, people, rights or opportunities must be protected?                           |
| 3 | <b>Prompt</b>                  | If AI is appropriate, give it a clear task, context, boundaries and output requirements.       |
| 4 | <b>Prove</b>                   | Verify consequential claims, sources, outputs, assumptions and limitations.                    |
| 5 | <b>Preserve Human Judgment</b> | Name the person with authority to review, correct, stop, override and make the final decision. |

## How the two connect

|                                                 |                                                           |                                                 |                                                                        |
|-------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------|------------------------------------------------------------------------|
| <b>Tell It</b><br>deepens into<br><b>Prompt</b> | <b>Limit It</b><br>deepens into<br><b>Pause · Protect</b> | <b>Check It</b><br>deepens into<br><b>Prove</b> | <b>People Decide</b><br>deepens into<br><b>Preserve Human Judgment</b> |
|-------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------|------------------------------------------------------------------------|

## Decision legend

|                                                                |                                                                                                                        |                                                                                      |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>GREEN — GO</b><br>Low-consequence use with ordinary review. | <b>YELLOW — REVISE / REVIEW</b><br>Uncertainty or meaningful consequences require stronger verification or safeguards. | <b>RED — STOP</b><br>AI should not make or replace the consequential human decision. |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|

**Context and consequences decide the color—not the tool.** The same AI tool can be GREEN for brainstorming poster headlines, YELLOW for summarizing research you will publish, and RED for recommending a student's discipline or an applicant's rejection.

## YOUR NINE-WEEK PATHWAY

# Portfolio roadmap

Understand → Evaluate → Analyze → Protect → Apply & Create. Keep every artifact; Weeks 7–9 turn them into your capstone.

| WEEK | FOCUS · COMPETENCY                                                  | PORTFOLIO ARTIFACT                                        | AMBASSADOR                                                                            | DONE                     |
|------|---------------------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------|--------------------------|
| 1    | <b>AI Around Us</b><br>Understand                                   | AI ecosystem map + baseline                               |    | <input type="checkbox"/> |
| 2    | <b>Safety, Ethics &amp; Responsible Use</b><br>Protect              | Responsible-use decision record + agreement               |    | <input type="checkbox"/> |
| 3    | <b>How Generative AI Works—and Fails</b><br>Understand · Evaluate   | Generative AI explanation + output autopsy                |    | <input type="checkbox"/> |
| 4    | <b>Prompting, Research &amp; Verification</b><br>Evaluate           | Verified research brief + AI-use disclosure               |    | <input type="checkbox"/> |
| 5    | <b>Bias, Fairness &amp; AI-Supported Decisions</b><br>Analyze       | Stakeholder / safeguard recommendation                    |   | <input type="checkbox"/> |
| 6    | <b>Media Literacy &amp; Synthetic Content</b><br>Evaluate · Protect | Synthetic-media verification record + disclosure label    |  | <input type="checkbox"/> |
| 7    | <b>Problem Discovery &amp; AI Fit</b><br>Analyze · Apply            | Problem statement + stakeholder map + AI Fit test         | All six                                                                               | <input type="checkbox"/> |
| 8    | <b>Responsible Solution Design</b><br>Protect · Apply               | AI role map + risk register + prototype                   | All six                                                                               | <input type="checkbox"/> |
| 9    | <b>Capstone Pitch &amp; Reflection</b><br>Apply · Create            | Capstone portfolio + pitch + reflection + post-assessment | All six                                                                               | <input type="checkbox"/> |

## How your work is assessed

Weekly artifacts are scored on evidence, systems thinking, responsible use, verification, decision quality and communication. The capstone uses a seven-criterion rubric you'll see in Week 9. Evidence and reasoning count more than polish.

## Before you submit anything

- Remove names and personal details.
- Name the source of every important claim.
- Disclose meaningful AI assistance.

## WEEK 4 · PORTFOLIO ARTIFACT

## Verified research brief

ARTIFACT: VERIFIED BRIEF

Research question: \_\_\_\_\_

1 Claim: \_\_\_\_\_

## SOURCE 1

Who · organization · date · evidence  
\_\_\_\_\_  
\_\_\_\_\_

## SOURCE 2 (independent)

Who · organization · date · evidence  
\_\_\_\_\_  
\_\_\_\_\_

STATUS

☐

Corroborated

☐

Contradicted

☐

Mixed

☐

Not enough evidence

2 Claim: \_\_\_\_\_

## SOURCE 1

Who · organization · date · evidence  
\_\_\_\_\_  
\_\_\_\_\_

## SOURCE 2 (independent)

Who · organization · date · evidence  
\_\_\_\_\_  
\_\_\_\_\_

STATUS

☐

Corroborated

☐

Contradicted

☐

Mixed

☐

Not enough evidence

## Lateral-reading moves I used

☐

Checked the author

☐

Checked the organization

☐

Checked the date

☐

Followed the evidence to its origin

☐

Found independent corroboration

☐

Compared a primary source

## What changed after lateral reading?

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## 7

WEEK 7 · ANALYZE · APPLY

## Human-Centered Problem Discovery &amp; AI Fit

## ESSENTIAL QUESTION

What problem is worth solving—and is AI actually the right tool?

## YOUR MISSION

Your capstone starts now. Find a real, non-sensitive need, prove it with stakeholder evidence, and decide honestly whether AI belongs in the solution.

## QUICK PICK Which statement names a problem?

“We need an AI chatbot.”

☐ Names a problem

“Families miss schedule changes because updates are scattered across three channels.”

☐ Names a problem

## Capstone challenge menu


☐ Accessible information

Make school wellness resources easier to find—no personal health data.


☐ Event operations & logistics

Setup checklists or equipment checkout for shop, theater or athletics.


☐ School communication

Announcements scattered across channels; study-planning tools.


☐ Environmental observation

Map litter or campus trees from public spaces; aviation career awareness.


☐ Club & community operations

Volunteer sign-ups, ticketing or a community-resource directory.


☐ Sports media & career content

Game-day highlights workflow; career explainers for 9th graders.

☐ Propose our own (teacher approval): \_\_\_\_\_

## Team roles — first names only

**Problem & stakeholder lead** · evidence, observations, problem statement

**Responsible design lead** · risk register, safeguards, remedy, stop rules

**Systems & AI Fit lead** · workflow map, AI vs. simpler options

**Evidence & communication lead** · sources, disclosure, prototype, pitch

## WEEK 7 · PORTFOLIO ARTIFACT

# Stakeholder map

ARTIFACT: STAKEHOLDER MAP

Use names of roles or groups—never private details. Note each group's needs, risks or questions.

## Primary users

Who would use the solution?

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## Decision-makers

Who can approve, change or stop it?

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## THE PROBLEM

In a few words

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## People affected

Who feels the results, even without using it?

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## People with expertise

Who knows this problem best?

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## People who may be excluded

Who might be unable to use, understand, challenge or benefit from a solution?

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## Evidence we gathered

☐ Observation

☐ Approved public information

☐ Fictionalized interview

☐ Teacher-provided data

☐ Survey with no personal data

☐ We collected only the minimum evidence needed and no sensitive personal data.

## Evidence-based problem statement

\_\_\_\_\_ needs a way to \_\_\_\_\_

because \_\_\_\_\_

while respecting \_\_\_\_\_.

Stakeholder · need · evidence/context · constraint

## WEEK 8 · PORTFOLIO ARTIFACT

# AI role map

ARTIFACT: AI ROLE MAP

Trace your solution from purpose to stop rule. Navy steps are the human authority points—they must name a role, not “someone.”

1

**Purpose & intended benefit** \_\_\_\_\_

2

**Minimum necessary input** \_\_\_\_\_

What data, and only that?

3

**AI task — if any** \_\_\_\_\_

Exactly what does AI do?

4

**Output & uncertainty** \_\_\_\_\_

What does it produce? How sure is it?

5

**Human reviewer** \_\_\_\_\_

Who checks, with what time and information?

6

**Final authority** \_\_\_\_\_

Who makes the call?

7

**Override & fallback** \_\_\_\_\_

How do people intervene? What if it fails?

8

**Appeal / remedy** \_\_\_\_\_

How can an affected person fix a wrong outcome?

9

**Monitoring & stop rule** \_\_\_\_\_

What do you watch? When do you stop?

**HUMAN CHECKPOINT****Could a person actually intervene?**

Human review only counts if the reviewer has the authority, information and time to change the outcome. A disclaimer is not a fallback.

## WEEK 8 · PORTFOLIO ARTIFACT

# Risk register

ARTIFACT: RISK REGISTER

Rate each risk, then plan to prevent it, catch it and respond. Every major risk needs an owner and a stop rule.

| RISK               | RATING                                                                                                                                                                 | PREVENT | DETECT / RESPOND | OWNER / STOP RULE |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|------------------|-------------------|
| Privacy / data     | Likely<br><input type="radio"/> L <input type="radio"/> M <input type="radio"/> H<br>Impact<br><input type="radio"/> L <input type="radio"/> M <input type="radio"/> H |         |                  |                   |
| Accuracy           | Likely<br><input type="radio"/> L <input type="radio"/> M <input type="radio"/> H<br>Impact<br><input type="radio"/> L <input type="radio"/> M <input type="radio"/> H |         |                  |                   |
| Bias / fairness    | Likely<br><input type="radio"/> L <input type="radio"/> M <input type="radio"/> H<br>Impact<br><input type="radio"/> L <input type="radio"/> M <input type="radio"/> H |         |                  |                   |
| Access / exclusion | Likely<br><input type="radio"/> L <input type="radio"/> M <input type="radio"/> H<br>Impact<br><input type="radio"/> L <input type="radio"/> M <input type="radio"/> H |         |                  |                   |
| Misuse / harm      | Likely<br><input type="radio"/> L <input type="radio"/> M <input type="radio"/> H<br>Impact<br><input type="radio"/> L <input type="radio"/> M <input type="radio"/> H |         |                  |                   |
| Accountability     | Likely<br><input type="radio"/> L <input type="radio"/> M <input type="radio"/> H<br>Impact<br><input type="radio"/> L <input type="radio"/> M <input type="radio"/> H |         |                  |                   |

YOUR CALL

Our highest-rated risk is \_\_\_\_\_. If our safeguards fail, we will

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## WEEK 8 · DESIGN QA

# PRE-PILOT STRESS TEST

BEFORE WEEK 9

Try to find the weakness before the real world does.

Design reviewed (team) \_\_\_\_\_

Reviewing team \_\_\_\_\_

**Reviewers:** inspect the other team's role map, risk register and prototype. Check every box where you find a weakness, and note where you found it.

An unsupported assumption

☐

A stakeholder who may be missing

☐

A privacy or data concern

☐

An accuracy or evidence problem

☐

A fairness or access problem

☐

Unclear human authority

☐

A failure scenario

☐

A missing fallback or stop rule

☐

An accessibility issue

☐**Reviewer's strongest concern**

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**Evidence or reason**

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**YOUR CALL**

Our response:

☐ KEEP☐ REVISE☐ STOP & REDESIGN**What we changed before Week 9**

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This stress test checks the **design** before the pitch. Week 9 peer feedback evaluates your **communication** and **final reasoning**.

## WEEK 9 · PITCH PLANNER

# Four- to five-minute pitch

Plan each section in key words—not a script. Assign a presenter to every section.

| SECTION · WHAT THE AUDIENCE NEEDS                                                                        | OUR KEY POINTS | PRESENTER |
|----------------------------------------------------------------------------------------------------------|----------------|-----------|
| <b>1 Need</b><br>Who has the problem, and what evidence supports it?                                     |                |           |
| <b>2 Alternatives &amp; AI Fit</b><br>Why this approach—and why not a simpler option?                    |                |           |
| <b>3 Solution &amp; prototype</b><br>What happens from input to action?<br>Where can a person intervene? |                |           |
| <b>4 Risks &amp; safeguards</b><br>How are privacy, accuracy, fairness, access and misuse addressed?     |                |           |
| <b>5 Human authority &amp; remedy</b><br>Who decides, who can override, and how can people correct harm? |                |           |
| <b>6 Limitations &amp; next test</b><br>What isn't proven? What would you test before a pilot?           |                |           |

## PREPARE FOR QUESTIONS

Check the two questions you're most likely to get. Draft one strong answer.

- ☐ What evidence changed your design?
 ☐ What could go wrong—and who bears the risk?
- ☐ What happens when the system is uncertain?
 ☐ Who can stop it?
- ☐ What would make you choose a non-AI option?

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